Software Test Plan (STP) - Hostel Management System

1. Introduction

Purpose: This document defines the test plan for the Hostel Management System . It sets out the objectives, scope, strategy, environment, schedule and responsibilities for testing, and contains the designed unit, integration and system test cases for every use case.

Scope: Testing covers login and role-based access, student management, hostel and room management, room availability, room allocation with over-allocation prevention, modify and cancel, allocation history, check-in and check-out, the admin dashboard, and the performance, reliability, security, usability, maintainability and data integrity requirements.

2. Test Items

- **AuthModule (accounts):** login, logout, sessions, @admin_required role check
- **StudentModule (students):** registration, update, own profile, search
- **HostelModule (hostels):** hostel create and update
- **RoomModule (rooms):** room create and update, capacity rules, available_capacity(), availability JSON endpoint
- **AllocationModule (allocations):** allocate_room() with over-allocation check, modify, cancel, AllocationHistory
- **Check-In and Check-Out Modules (allocations):** check_in(), check_out()
- **DashboardModule (dashboard):** administrative summary
- **Web UI:** admin pages and student home page (Django templates)

3. Features to be Tested

Requirement IDs	Feature	Use case in Appendix A
HMS-F-001, F-002, SR-001, SR-002, SR-005	Login, role-based access, session end on logout	UC-01
HMS-F-003, F-004, F-005, F-017, SR-004, SR-006	Register, update, view own profile, search students	UC-02
HMS-F-006, F-007, F-009	Hostel and room management, room details	UC-03
HMS-F-008, NF-006	Room availability calculation and display	UC-04
HMS-F-010, F-011	Room allocation, prevent over-allocation	UC-05
HMS-F-012, F-013, F-014, F-018	View status, modify, cancel, allocation history	UC-06
HMS-F-015, F-016	Check-in and check-out	UC-07
HMS-F-019	Administrative summary	UC-08
HMS-NF-001	Pages within 3 s for 90% of requests	NFR set
HMS-NF-002 to NF-006, SR-002 to SR-006	Reliability, security, usability, maintainability, data integrity	NFR set

4. Features Not to be Tested

- Out-of-scope features in the SRS: online fee payment, biometric attendance, visitor management, integration with other college systems.
- Internal behaviour of third-party components: Django, the ORM, SQLite / PostgreSQL, Gunicorn, Nginx (assumed tested by their maintainers; only our use of them is tested).
- Browsers other than the latest Google Chrome and Mozilla Firefox.
- Load beyond normal academic project load (thousands of simultaneous users).
- Password reset by email and notifications (not part of this version).

5. Test Approach / Strategy

Levels:

- **Unit tests (UT):** Django TestCase on each module's model methods, forms and service functions (for example available_capacity(), allocate_room(), check_out()), run on an in-memory SQLite test database. Target: 80% line coverage on models and services, reported by coverage.py.
- **Integration tests (IT):** Django test Client requests through URLs, views, forms and the database together, including login and role checks.
- **System tests (ST):** end-to-end flows in the browser with seeded data, manual first and automated with Selenium for login, allocation and check-out.
- **Acceptance tests (UAT):** the course coordinator watches the ST scenarios in a demo and signs off against the SRS acceptance criteria.

Types:

- Functional testing of every SRS requirement with valid and invalid inputs.
- Boundary value testing: room capacity 0, 1 and full; last free bed; duplicate SRN and room number.
- State transition testing of the allocation life cycle: ACTIVE to COMPLETED or CANCELLED; check-out before check-in must fail.
- Regression testing: the full UT and IT suite (python manage.py test) is run before every merge to main and after every bug fix.
- Performance testing with Locust against HMS-NF-001.
- Security testing (section 5.1).
- Usability testing with a short task-based walk-through.

Defect severity levels:

Severity	**Meaning**	**Example**
Critical	Wrong data or security hole; blocks testing	Room gets more allocations than capacity; student opens admin pages
High	Main feature broken, no workaround	Check-out does not free the bed
Medium	Feature partly works or has a workaround	Search works by SRN but not by name
Low	Cosmetic or text issue	Spelling mistake, misaligned button

Entry criteria: feature merged to main with its unit tests; migrations run cleanly; seed data available; test cases for the feature reviewed.

Exit criteria: 100% of planned test cases executed; all High priority requirements passed; 0 open Critical or High defects; coverage target met; RTM shows every requirement passed.

5.1 Security Validation

- **Authentication:** valid and invalid login, inactive accounts, session ends on logout (UC-01).
- **Authorisation:** @admin_required on every admin URL and object-level checks so students only see their own data; manual IDOR test with each role (IT_AUTH_03, IT_STU_03, ST_SEC_02).
- **Passwords:** stored with Django's PBKDF2 hasher, never shown or logged (ST_SEC_01).
- **Input handling:** form validation, SQL injection and XSS strings in all text fields and search (UT_STU_03, ST_SEC_04).
- **CSRF:** POST forms submitted without a CSRF token must be rejected (checked as part of ST_SEC_04).
- **Configuration:** DEBUG=False, secret key outside the repository, secure cookies on HTTPS (checked with python manage.py check --deploy before the demo).

6. Test Environment

Item	Details
Hardware	Team laptops (8 GB RAM, Intel i5 / Ryzen 5 or better)
Software	Windows 11 and Ubuntu 22.04, Python 3.11+, Django 5.x, SQLite 3; Gunicorn + Nginx for the demo deployment
Browsers	Google Chrome and Mozilla Firefox (latest)
Tools	Django test framework and test Client, coverage.py, Selenium WebDriver, Locust, DB Browser for SQLite, curl, GitHub Issues (defects)
Test data	Seed script creating 1 admin, 3 student logins, 2 hostels, 12 rooms (capacities 1 to 3, some full) and 25 students; a larger script with 500 students and 100 rooms for ST_PERF_01. Dummy data only.

7. Test Schedule

Milestone	Date
Test plan and test case design	23-10-2026 to 26-10-2026
Test environment setup and seed data scripts	27-10-2026
Testing cycle 1: UC-01, UC-02, UC-03, UC-04	28-10-2026 to 01-11-2026
Testing cycle 2: UC-05, UC-06, UC-07, UC-08	02-11-2026 to 06-11-2026
Performance and security test cycle (NFR set)	07-11-2026 to 09-11-2026
Defect fixing, retesting and regression	09-11-2026 to 11-11-2026
UAT / demo	12-11-2026 to 13-11-2026

Milestone	Date
Test summary report	14-11-2026

8. Test Deliverables

- Test Plan (this document)
- Test cases (Appendix A) and automated tests in the repository (tests.py in each Django app; each test name contains its test case ID)
- Test data seed scripts
- Test execution logs and coverage.py report
- Defect reports in GitHub Issues, linked to the failing test case
- Locust performance report
- Test Summary Report and updated RTM

9. Risks and Mitigation

Risk	Mitigation
Features finish late, leaving little time to test	Unit tests written with each feature; a feature is done only when its tests pass
Members busy with other course deadlines	Fixed testing cycles in the schedule; each module has a backup tester
Developers testing their own code miss their own blind spots	Cross-testing: system tests for a module are run by a different member
Concurrency bug (double allocation) hard to reproduce manually	Automated UT_ALLOC_05 with two threads, plus review of the transaction code
Test data changes between runs and causes false failures	Reload seed data before every system test cycle; unit tests use a fresh test database
Performance results differ between laptops	Run ST_PERF_01 on one fixed machine with DEBUG=False and record its specs

10. Suspension & Resumption Criteria

Suspend testing if: the build does not start or migrations fail; the build is unstable and blocks more than 30% of test cases; a Critical security or data integrity defect is found (stop and fix before continuing).

Resume testing when: blocking defects are fixed and verified; seed data reloaded; a smoke run of ST_AUTH_01 and ST_ALLOC_01 passes.

11. Test Case Management & Traceability

Test cases live in Appendix A and as automated tests in the repository; each automated test name contains its test case ID (for example test_UT_ALLOC_03_full_room_rejected). Defects are logged in GitHub Issues and linked to the failing test case. The table below extends the SRS RTM with the designed test cases.

Requirement	SRS test ref	Designed test cases
HMS-F-001	TC-AUTH-01	UT_AUTH_01, UT_AUTH_02, ST_AUTH_01
HMS-F-002	TC-AUTH-02	UT_AUTH_04, IT_AUTH_03, IT_CO_02
HMS-F-003	TC-STU-01	UT_STU_01, UT_STU_02, ST_STU_01
HMS-F-004	TC-STU-02	UT_STU_04, ST_STU_01
HMS-F-005	TC-STU-03	IT_STU_02
HMS-F-006	TC-HOSTEL-01	UT_HOS_01, ST_ROOM_01
HMS-F-007	TC-ROOM-01	UT_ROOM_01, UT_ROOM_02, UT_ROOM_03, ST_ROOM_01
HMS-F-008	TC-ROOM-02	UT_AVL_02, UT_AVL_03, IT_AVL_01, ST_AVL_01
HMS-F-009	TC-ROOM-03	IT_ROOM_02
HMS-F-010	TC-ALLOC-01	UT_ALLOC_01, UT_ALLOC_02, IT_ALLOC_01, ST_ALLOC_01
HMS-F-011	TC-ALLOC-02	UT_ALLOC_03, UT_ALLOC_04, UT_ALLOC_05, IT_ALLOC_02, ST_ALLOC_01
HMS-F-012	TC-ALLOC-03	IT_STAT_01
HMS-F-013	TC-ALLOC-04	UT_MOD_01, UT_MOD_02, ST_ALLOC_02
HMS-F-014	TC-ALLOC-05	UT_CAN_01, ST_ALLOC_02
HMS-F-015	TC-CI-01	UT_CI_01, UT_CI_02, ST_CICO_01
HMS-F-016	TC-CO-01	UT_CO_01, UT_CO_02, IT_CO_01, ST_CICO_01
HMS-F-017	TC-STU-04	UT_STU_05, ST_STU_01
HMS-F-018	TC-ALLOC-06	UT_HIS_01, IT_HIS_01, ST_ALLOC_02
HMS-F-019	TC-DASH-01	UT_DASH_01, IT_DASH_01, ST_DASH_01
HMS-NF-001	TC-PERF-01	ST_PERF_01
HMS-NF-002	TC-REL-01	ST_REL_01
HMS-NF-003, SR-001	TC-SEC-01	IT_AUTH_02
HMS-SR-002	TC-SEC-02	UT_AUTH_04, IT_AUTH_03, ST_SEC_02
HMS-SR-003	TC-SEC-03	ST_SEC_01
HMS-SR-004	TC-SEC-04	UT_STU_03, ST_SEC_04
HMS-SR-005	TC-SEC-05	IT_AUTH_04
HMS-SR-006	TC-SEC-06	IT_STU_03, ST_SEC_02
HMS-NF-004	TC-UX-01	ST_UX_01
HMS-NF-005	TC-MAINT-01	ST_MAINT_01
HMS-NF-006	TC-DATA-01	UT_ALLOC_05, UT_AVL_03, ST_DATA_01

12. Test Metrics & Reporting

Planned test cases: 58 in total (28 unit, 14 integration, 16 system).

- Percentage of test cases executed and passed / failed, per use case
- Line coverage of models and services per module (coverage.py report)
- Defect count by severity, defect density per module, defect age (days open)
- Requirement coverage from the RTM (requirements with all designed cases passed / total requirements)

Reporting: status shared in the team group at the end of each testing day; RTM in the repository updated weekly; a Test Summary Report at the end of the cycle.

Appendix A: Test Cases

ID convention: UT = unit test, IT = integration test, ST = system test, followed by the module code and number (for example UT_ALLOC_01). Actual Result and Test Result are filled in during execution; cases are listed here as designed and not yet executed.

UC-01 Login and Access Control (HMS-F-001, F-002, SR-001, SR-002, SR-005)

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_AUTH_01	AuthModule / authenticate()	Valid credentials return the user (HMS-F-001)	Test DB with seeded admin user	1: Call authenticate(username, password) 2: Inspect returned user	admin1 / Test@1234	Returns User object with is_staff = True	Not yet executed	Pending
UT_AUTH_02	AuthModule / authenticate()	Wrong password is rejected	Seeded admin user	1: Call authenticate() with wrong password	admin1 / wrong123	Returns None; no session created	Not yet executed	Pending
UT_AUTH_04	AuthModule / admin_required	Decorator blocks non-admin users (HMS-F-002)	RequestFactory requests for a student user and an admin user	1: Call a view wrapped with @admin_required as student 2: Repeat as admin	stu1, admin1	Step 1 raises PermissionDenied (403); step 2 returns 200	Not yet executed	Pending
IT_AUTH_02	URLs / login_required	Protected pages need login (HMS-SR-001)	Not logged in	1: GET /students/, /rooms/availability/, /student/home/	-	Each returns 302 to /accounts/login/?next=<url>	Not yet executed	Pending
IT_AUTH_03	URLs / admin_required	Student session cannot reach admin pages (HMS-F-002, SR-002)	Logged in as stu1	1: GET /students/new/, /rooms/new/, /allocations/new/ 2: POST to /allocations/new/	Valid allocation form data	HTTP 403 for every request; no allocation saved	Not yet executed	Pending
IT_AUTH_04	AuthModule / logout	Logout ends the session (HMS-SR-005)	Logged in as stu1	1: POST /accounts/logout/ 2: GET /student/home/ with the old session cookie	Session cookie from login	Step 2 returns 302 to login page	Not yet executed	Pending
ST_AUTH_01	Login page (UI)	End-to-end login lands on the right page for each role	Chrome browser; seeded admin and student accounts	1: Open /accounts/login/ 2: Enter username and password 3: Click Login 4: Repeat for each role and once with a wrong password	admin1, stu1 / Test@1234; wrong123	Admin sees dashboard, student sees own home page; wrong password shows 'Invalid username or password'	Not yet executed	Pending

UC-02 Student Management (HMS-F-003, F-004, F-005, F-017, SR-004, SR-006)

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_STU_01	StudentModule / StudentForm	Valid student details are saved (HMS-F-003)	Empty test DB	1: Validate and save StudentForm 2: Query Student by SRN	SRN: PES1UG24AM999, Name: Test Student, Email: test@pes.edu, Phone: 9876543210, Year: 2	Form valid; one Student row with these values	Not yet executed	Pending

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_STU_02	StudentModule / StudentForm	Duplicate SRN is rejected	Student PES1UG24AM999 exists	1: Validate StudentForm with the same SRN	SRN: PES1UG24AM999	Form invalid; error 'Student with this SRN already exists.'	Not yet executed	Pending
UT_STU_03	StudentModule / StudentForm	Invalid email and phone are rejected (HMS-SR-004)	None	1: Validate StudentForm	Email: abc, Phone: 98ab12	Form invalid; errors on email and phone fields	Not yet executed	Pending
UT_STU_04	StudentModule / update	Updated details are saved (HMS-F-004)	Student PES1UG24AM999 exists	1: Save StudentForm(instance) with new phone and year 2: Reload from DB	Phone: 9123456780, Year: 3	Reloaded record has new phone and year	Not yet executed	Pending
UT_STU_05	StudentModule / search_students()	Search by SRN or part of name, case-insensitive (HMS-F-017)	Students 'Riya Rao' (AM101) and 'Rahul Shah' (AM102)	1: search_students('AM101') 2: search_students('riya') 3: search_students('zzzz')	Queries as listed	1: Riya Rao; 2: Riya Rao; 3: empty result	Not yet executed	Pending
IT_STU_02	URLs / student home	Student sees only own profile (HMS-F-005)	stu1 linked to Riya Rao	1: Login as stu1 2: GET /student/home/	stu1 / Test@1234	200; page shows Riya Rao's details only	Not yet executed	Pending
IT_STU_03	URLs / object access	Student cannot open another student's record (HMS-SR-006)	stu1 and stu2 exist	1: Login as stu1 2: GET /students/<stu2 id>/edit/	stu2 id	HTTP 403; no data of stu2 in response	Not yet executed	Pending
ST_STU_01	Students pages (UI)	Admin registers, edits and searches a student	Admin logged in in Chrome	1: Students > Add, fill form, Save 2: Edit phone, Save 3: Search by name	Same as UT_STU_01	Success messages shown; updated phone visible; search finds the student	Not yet executed	Pending

UC-03 Hostel and Room Management (HMS-F-006, F-007, F-009)

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_HOS_01	HostelModule / HostelForm	Valid hostel is saved (HMS-F-006)	Empty test DB	1: Save HostelForm	Name: Block C, Type: Boys, Warden: Mr. Rao	Hostel row created	Not yet executed	Pending
UT_ROOM_01	RoomModule / RoomForm	Valid room is saved (HMS-F-007)	Hostel Block C exists	1: Save RoomForm	Room: C-101, Floor: 1, Capacity: 3	Room row linked to Block C	Not yet executed	Pending
UT_ROOM_02	RoomModule / capacity	Capacity boundary values	Hostel Block C exists	1: Validate capacity 0 2: Validate capacity -1 3: Validate capacity 1	Capacity: 0, -1, 1	0 and -1 invalid ('Capacity must be at least 1'); 1 valid	Not yet executed	Pending
UT_ROOM_03	RoomModule / uniqueness	Room number unique within a hostel	C-101 exists in Block C; Block D exists	1: Create C-101 in Block C 2: Create C-101 in Block D	Room: C-101	Step 1 rejected; step 2 allowed	Not yet executed	Pending
IT_ROOM_02	URLs / room details	Room details visible to admin and student (HMS-F-009)	Room C-101 exists	1: GET /rooms/<id>/ as admin1 2: GET /rooms/<id>/ as stu1	-	200 for both; hostel, number, capacity, occupied and available shown	Not yet executed	Pending
ST_ROOM_01	Hostels and Rooms pages (UI)	Admin sets up a hostel with rooms	Admin logged in	1: Add hostel 2: Add 3 rooms with capacities 1, 2,	Block E; E-1, E-2, E-3	All rooms listed with correct capacity and availability	Not yet executed	Pending

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
				3 3: Open room list				

UC-04 Room Availability (HMS-F-008, NF-006)

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_AVL_02	RoomModule / available_capacity()	Active allocations reduce availability	Room capacity 3 with 2 ACTIVE allocations	1: Call room.available_capacity()	-	Returns 1	Not yet executed	Pending
UT_AVL_03	RoomModule / available_capacity()	Cancelled and completed allocations are not counted	Room capacity 3: 1 ACTIVE, 1 CANCELLED, 1 COMPLETED	1: Call room.available_capacity()	-	Returns 2	Not yet executed	Pending
IT_AVL_01	JSON API / availability	Availability endpoint returns correct values	Room A-101 capacity 3, 2 ACTIVE; logged in as stu1	1: GET /api/rooms/<id>/availability/	-	200 {"capacity": 3, "active_allocations": 2, "available": 1}	Not yet executed	Pending
ST_AVL_01	Availability page (UI)	Student view updates after admin allocates	Admin and student logged in (two browsers)	1: Student notes availability of A-101 2: Admin allocates a student to A-101 3: Student refreshes	Room A-101	Available beds reduced by 1 after refresh	Not yet executed	Pending

UC-05 Room Allocation (HMS-F-010, F-011)

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_ALLOC_01	AllocationModule / allocate_room()	Allocation succeeds when a bed is free (HMS-F-010)	Student without allocation; room capacity 2, empty	1: Call allocate_room(student, room, admin)	-	Allocation with status ACTIVE created; one history row CREATED; availability 1	Not yet executed	Pending
UT_ALLOC_02	AllocationModule / allocate_room()	Student with an active allocation is rejected	Student already ACTIVE in another room	1: Call allocate_room() again	-	Raises AlreadyAllocatedError; no new rows	Not yet executed	Pending
UT_ALLOC_03	AllocationModule / allocate_room()	Full room is rejected (HMS-F-011)	Room capacity 2 with 2 ACTIVE allocations	1: Call allocate_room() for a third student	-	Raises RoomFullError; active count stays 2	Not yet executed	Pending
UT_ALLOC_04	AllocationModule / allocate_room()	Boundary: last bed	Room capacity 1, empty; students A and B	1: Allocate A 2: Allocate B	-	Step 1 succeeds; step 2 raises RoomFullError	Not yet executed	Pending
UT_ALLOC_05	AllocationModule / concurrency	Two requests for the last bed at the same time	Room with 1 free bed; TransactionTestCase	1: Start two threads calling allocate_room() for different students 2: Count ACTIVE allocations	-	Exactly one succeeds, the other raises RoomFullError; active count = capacity	Not yet executed	Pending

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
IT_ALLOC_01	URLs + AllocationModule	Allocate through the view	Logged in as admin1; free room and unallocated student	1: POST /allocations/new/	student_id, room_id	302 with message 'Room allocated successfully'	Not yet executed	Pending
IT_ALLOC_02	URLs + AllocationModule	Allocate to a full room through the view	Room full	1: POST /allocations/new/	student_id, full room_id	200 form with error 'Room has reached its capacity'; no allocation saved	Not yet executed	Pending
ST_ALLOC_01	Allocate page (UI)	Admin fills a room until it is full	Admin logged in; room capacity 2	1: Allocate 2 students to the room 2: Try a third using a stale browser tab	Room E-2	First two succeed; third shows capacity error; room shows 'Full'	Not yet executed	Pending

UC-06 View, Modify and Cancel Allocation, History (HMS-F-012, F-013, F-014, F-018)

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_MOD_01	AllocationModule / modify_allocation()	Move student to a room with a free bed (HMS-F-013)	Student ACTIVE in room A; room B has a free bed	1: Call modify_allocation(alloc, B, admin)	-	Allocation room = B; A availability +1, B availability -1; history MODIFIED with old room A, new room B	Not yet executed	Pending
UT_MOD_02	AllocationModule / modify_allocation()	Move to a full room is rejected	Room B full	1: Call modify_allocation(alloc, B, admin)	-	Raises RoomFullError; student stays in room A	Not yet executed	Pending
UT_CAN_01	AllocationModule / cancel_allocation()	Cancel frees the bed (HMS-F-014)	ACTIVE allocation	1: Call cancel_allocation(alloc, admin)	-	Status CANCELLED; history CANCELLED; room availability +1	Not yet executed	Pending
UT_HIS_01	AllocationModule / history	History records every action (HMS-F-018)	Allocation created, modified and cancelled by admin1	1: Query AllocationHistory for the allocation	-	3 rows in order CREATED, MODIFIED, CANCELLED, each with performed_by = admin1 and timestamp	Not yet executed	Pending
IT_STAT_01	URLs / student home	Student sees allocation status (HMS-F-012)	stu1 has ACTIVE allocation; stu2 has none	1: GET /student/home/ as stu1 2: GET /student/home/ as stu2	-	1: hostel, room and status shown; 2: 'No room allocated yet'	Not yet executed	Pending
IT_HIS_01	URLs / history	History page filter and access	History rows exist	1: GET /allocations/history/?student=<id> as admin1 2: Same as stu1	-	1: matching rows listed; 2: 403	Not yet executed	Pending
ST_ALLOC_02	Allocation pages (UI)	Admin modifies, cancels and checks history	Admin logged in; ACTIVE allocation	1: Modify room 2: Cancel with confirmation 3: Open history	-	Messages shown for each step; history shows all three actions	Not yet executed	Pending

UC-07 Check-in and Check-out (HMS-F-015, F-016)

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_CI_01	AllocationModule / check_in()	Check-in an allocated student (HMS-F-015)	ACTIVE allocation, not checked in	1: Call check_in(alloc, admin)	-	checked_in_on set to now; history CHECKED_IN	Not yet executed	Pending
UT_CI_02	AllocationModule / check_in()	Check-in twice is rejected	Already checked in	1: Call check_in() again	-	Raises InvalidStateError; date unchanged	Not yet executed	Pending
UT_CO_01	AllocationModule / check_out()	Check-out frees the bed (HMS-F-016)	ACTIVE allocation, checked in	1: Call check_out(alloc, admin)	-	Status COMPLETED; checked_out_on set; history CHECKED_OUT; room availability +1	Not yet executed	Pending
UT_CO_02	AllocationModule / check_out()	Check-out before check-in is rejected	ACTIVE allocation, not checked in	1: Call check_out()	-	Raises InvalidStateError ('Student is not checked in')	Not yet executed	Pending
IT_CO_01	URLs + Check-Out Module	Check-out updates availability	Checked-in allocation in room with 0 free beds	1: POST /allocations/<id>/checkout/ 2: GET /api/rooms/<id>/availability/	-	Step 1: 302 with message; step 2: available = 1	Not yet executed	Pending
IT_CO_02	URLs / admin_required	Student cannot check out	Logged in as stu1	1: POST /allocations/<id>/checkout/	-	403; allocation unchanged	Not yet executed	Pending
ST_CICO_01	Allocation detail (UI)	Full stay: allocate, check-in, check-out	Admin logged in	1: Allocate student 2: Click Check-in 3: Click Check-out and confirm	-	Buttons enabled only when allowed; final status COMPLETED; bed free again	Not yet executed	Pending

UC-08 Administrative Summary (HMS-F-019)

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT_DASH_01	DashboardModule / get_summary()	Summary counts match data	Seed data: 25 students, 12 rooms, 24 beds, 15 ACTIVE allocations	1: Call get_summary()	-	students 25, rooms 12, beds 24, occupied 15, free 9	Not yet executed	Pending
IT_DASH_01	URLs / dashboard	Dashboard access	Seed data	1: GET /dashboard/ as admin1 2: GET /dashboard/ as stu1	-	1: 200 with values from UT_DASH_01; 2: 403	Not yet executed	Pending
ST_DASH_01	Dashboard (UI)	Dashboard matches the database	Admin logged in; seed data	1: Open dashboard 2: Compare with counts in DB Browser for SQLite	-	All values match	Not yet executed	Pending

Non-functional and Security Test Cases (HMS-NF-001 to NF-006, SR-002, SR-003, SR-004, SR-006)

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
ST_PERF_01	Whole system	90% of pages within 3 s (HMS-NF-001)	DEBUG=False; 500 students, 100 rooms seeded	1: Run Locust with 20 users for 5 minutes on list, availability and allocate pages 2: Read the 90th percentile	Locust script	90th percentile response time 3 s or less; 0 server errors	Not yet executed	Pending
ST_REL_01	Whole system / database	Data survives restart (HMS-NF-002)	Seed data plus new records	1: Create student and allocation 2: Stop and restart server 3: Check records	-	All records present and unchanged	Not yet executed	Pending
ST_SEC_01	AuthModule / storage	Passwords stored hashed (HMS-SR-003)	Users exist; log level INFO	1: Inspect auth_user.password in DB Browser 2: Search logs/hms.log for passwords	-	Values start with pbkdf2_sha256$; no plaintext password in DB or log	Not yet executed	Pending
ST_SEC_02	All URLs (manual IDOR test)	Broken access control check (HMS-SR-002, SR-006)	Accounts admin1, stu1, stu2	1: As stu1, replay every admin URL and stu2's IDs from the route list	Route list from SAD 3.3	Every attempt returns 403 or 404; no foreign data shown	Not yet executed	Pending
ST_SEC_04	Forms and search (fuzzing)	Injection, bad input and CSRF (HMS-SR-004)	Logged in as admin1	1: Enter ' OR 1=1 -- , <script> tags and 500 character strings in all text fields and search 2: Submit a form with curl without the CSRF token	Strings as listed	Rejected or stored as plain text; no extra records returned; no 500 error; step 2 returns 403	Not yet executed	Pending
ST_UX_01	UI	Usability walk-through (HMS-NF-004)	3 students who did not build HMS	1: Ask each to register a student, allocate a room and check out a student	-	All tasks finished without help; users rate messages as clear	Not yet executed	Pending
ST_MAINT_01	Code structure	Modular structure (HMS-NF-005)	Source code on main branch	1: Review: one Django app per module, no raw SQL, no cross-app view imports 2: Change dashboard app, run full test suite	-	Checklist passes; all other apps' tests still pass	Not yet executed	Pending
ST_DATA_01	Room + Allocation Modules	Capacity and availability stay consistent (HMS-NF-006)	Seed data	1: Run script doing 200 random allocate, modify, cancel and check-out actions 2: Check every room	Random seed 42	For every room: ACTIVE count <= capacity and displayed availability = capacity - ACTIVE	Not yet executed	Pending